# The Generalized Expenditure Manifold

A Unified Geometric Framework for
Demand, Production, Monetary Dynamics, Political Economy, and Strategic State Capacity

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**Abstract**

This paper develops a generalized theory of expenditure based upon a nonlinear manifold linking demand, price, production, technology, institutions, and state capacity.

The central object of analysis is the Generalized Expenditure Manifold

$$PD = AK^{\alpha}L^{\beta}T^{\gamma},$$

where $P$ denotes price, $D$ denotes demand, $K$ denotes capital, $L$ denotes labor, $T$ denotes technology, and $A$ denotes institutional efficiency.

The framework generalizes traditional demand theory by replacing the canonical linear demand curve with a family of hyperbolic expenditure surfaces generated by productive capacity.

Within this framework:

1. Monetary policy alters expenditure capacity.
2. Fiscal policy alters feasible regions.
3. Institutions determine manifold efficiency.
4. Technology determines manifold curvature.
5. Warfare reshapes the manifold through destruction and innovation.
6. Geopolitical power influences attainable expenditure capacity.
7. Classical linear demand curves emerge as local tangent approximations.

The resulting framework unifies concepts from economics, finance, psychology, philosophy, political science, international relations, warfare economics, statistics, computer science, and artificial intelligence.

The paper ends with "The End"

## Contents

# 1 Introduction

Modern economics traditionally represents demand by a linear schedule

$$P = a - bD.$$

Such specifications possess considerable computational convenience.
However, linear demand curves suffer from several weaknesses:

1. They imply variable elasticity.
2. They may predict negative prices.
3. They may predict negative quantities.
4. They possess no obvious micro-foundation linking them to productive capacity.
5. They fail to explicitly incorporate institutions, technology, and state capacity.

This paper proposes an alternative starting point.
Instead of assuming a demand schedule, we begin with an expenditure identity:

$$PD = B.$$

The quantity $B$ represents aggregate expenditure capacity.
The classical hyperbola follows immediately:

$$P = \frac{B}{D}.$$

We then generalize expenditure capacity itself as a function of productive resources:

$$B = AK^{\alpha}L^{\beta}T^{\gamma}.$$

Substitution yields the Generalized Expenditure Manifold:

$$PD = AK^{\alpha}L^{\beta}T^{\gamma}.$$

This equation forms the foundation of the present theory.

## 2 Motivation

The standard demand curve treats demand as primary.
The present framework treats productive capacity as primary.
The causal sequence becomes

$$(K, L, T, A) \to B \to (P, D).$$

The resulting geometry emerges naturally from production rather than being imposed externally.

## 3 The Fundamental Expenditure Identity

**Definition 1** (Expenditure Capacity)**.** *Expenditure capacity is defined by*

$$B > 0.$$

*The expenditure identity is*

$$PD = B.$$

Solving for price:

$$P = \frac{B}{D}.$$

Likewise,

$$D = \frac{B}{P}.$$

The resulting curve is a rectangular hyperbola.

**Proposition 1.** *The expenditure identity generates a rectangular hyperbola in the $(D, P)$ plane.*

*Proof.* Substitution gives

$$P = \frac{B}{D}.$$

Multiplying both sides by $D$ yields

$$PD = B.$$

This is the standard equation of a rectangular hyperbola.

□

# 4 The Generalized Production Function

We define expenditure capacity as

$$B = AK^{\alpha}L^{\beta}T^{\gamma}.$$

where

- $K$ = capital,
- $L$ = labor,
- $T$ = technology,
- $A$ = institutional efficiency.

Substitution into the expenditure identity yields

$$PD = AK^{\alpha}L^{\beta}T^{\gamma}.$$

This is the Generalized Expenditure Manifold.

**Definition 2** (Generalized Expenditure Manifold)**.** *The manifold is*

$$\mathcal{M} = (P, D, K, L, T, A) : PD = AK^{\alpha}L^{\beta}T^{\gamma}.$$

# 5 Logarithmic Representation

Taking logarithms:

$$\ln P + \ln D = \ln A + \alpha \ln K + \beta \ln L + \gamma \ln T.$$

The model therefore admits straightforward econometric estimation.

# 6 Existence Theorem

**Theorem 1** (Existence of the Manifold)**.** *Suppose*

$$A > 0, \quad K > 0, \quad L > 0, \quad T > 0.$$

*Then there exists a non-empty expenditure manifold satisfying*

$$PD = AK^{\alpha}L^{\beta}T^{\gamma}.$$

*Proof.* Define

$$B = AK^{\alpha}L^{\beta}T^{\gamma}.$$

Because every factor is positive,

$$B > 0.$$

Choose any

$$D > 0.$$

Then define

$$P = \frac{B}{D}.$$

Hence

$$PD = B.$$

The manifold is therefore non-empty. □

# 7 Geometry of the Manifold

Fix

$$(K, L, T, A).$$

Then

$$PD = B$$

is a single hyperbola.
If productive factors vary continuously, a family of hyperbolae emerges.
The resulting structure is a manifold embedded in a higher-dimensional state space.

## 7.1 A Cross Section

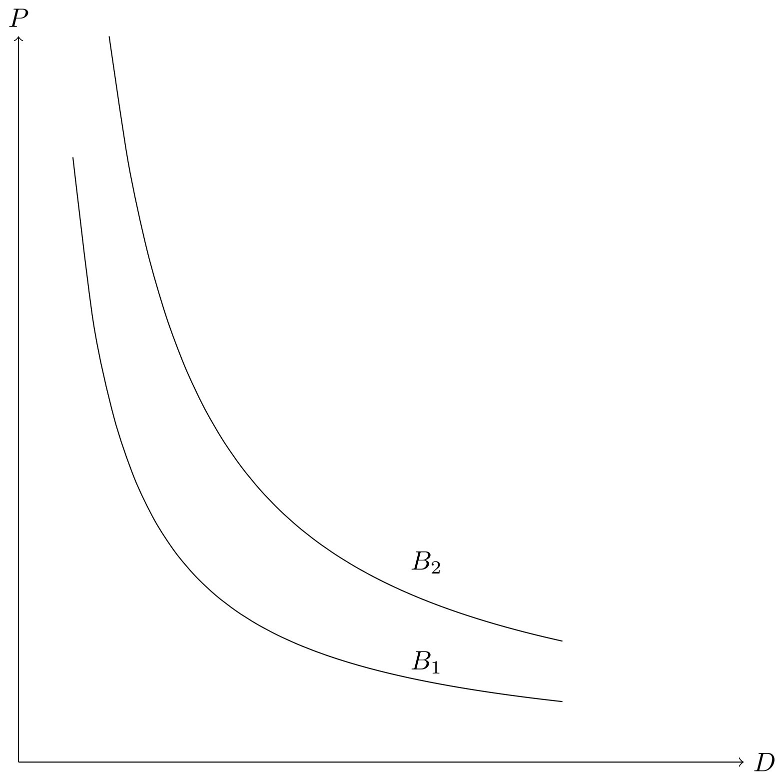


Figure 1: Two expenditure hyperbolae corresponding to different productive capacities.

# 8 Monetary Policy

Monetary policy changes nominal expenditure capacity.
Let

$$B = B(M, r, \pi_e),$$

where

- $M$ = money supply,
- $r$ = interest rate,
- $\pi_e$ = expected inflation.

Expansionary monetary policy increases

$$B.$$

Consequently

$$PD = B$$

moves outward.

**Proposition 2.** *Monetary policy changes the scale parameter of the expenditure manifold.*

*Proof.* The manifold equation remains

$$PD = AK^{\alpha}L^{\beta}T^{\gamma}.$$

Only the effective expenditure level changes.
The geometry remains hyperbolic.

□

# 9 Fiscal Policy

Fiscal policy affects:

1. taxes,
2. subsidies,
3. transfers,
4. infrastructure,
5. regulation.

Define effective ownership cost

$$C_O = P + R(P),$$

where $R(P)$ denotes repair and maintenance costs.
Fiscal policy affects

$$R(P).$$

Hence fiscal policy determines which regions of the manifold are economically feasible.

# 10 Boundary Equilibria

Suppose

$$D_{\min} \leq D \leq D_{\max}.$$

Then

$$P_{\max} = \frac{B}{D_{\min}},$$

and

$$P_{\min} = \frac{B}{D_{\max}}.$$

The feasible arc contains two boundary equilibria:

$$(P_{\max}, D_{\min}),$$

and

$$(P_{\min}, D_{\max}).$$

### 10.1 The High-Demand Boundary

The point

$$(P_{\min}, D_{\max})$$

represents maximal market penetration.
It emerges when value-for-money is highest.

### 10.2 The High-Price Boundary

The point

$$(P_{\max}, D_{\min})$$

represents scarcity, exclusivity, or extreme ownership cost.
Repair costs, maintenance costs, and institutional barriers often become dominant near this boundary.

## 11 The Tangent Approximation

Consider

$$P(D) = \frac{B}{D}.$$

Let

$$D_0 > 0.$$

Differentiation yields

$$\frac{dP}{dD} = -\frac{B}{D^2}.$$

Evaluated at $D_0$:

$$\left.\frac{dP}{dD}\right|_{D_0} = -\frac{P_0}{D_0}.$$

The first-order Taylor expansion becomes

$$P \approx P_0 - \frac{P_0}{D_0}(D - D_0).$$

This simplifies to

$$P = a - bD.$$

**Theorem 2** (Tangent Demand Theorem)**.** *The canonical linear demand curve is the first-order Taylor approximation of the expenditure manifold.*

*Proof.* Apply Taylor expansion around $(P_0, D_0)$.
The resulting approximation is linear. □

## 12 Elasticity

For

$$PD = B,$$

we obtain

$$D = \frac{B}{P}.$$

Differentiating:

$$\frac{dD}{dP} = -\frac{B}{P^2}.$$

Elasticity becomes

$$\varepsilon = \frac{dD}{dP}\frac{P}{D} = -1.$$

Thus hyperbolic demand possesses constant elasticity.
Linear demand does not.

# 13 Philosophical Foundations

The Generalized Expenditure Manifold suggests a philosophical inversion of traditional economics.

Classical approaches assume demand determines production.

The present framework assumes production determines expenditure capacity, which subsequently determines attainable demand.

The manifold therefore represents a theory of economic possibility rather than merely a theory of consumer choice.

# 14 Finance and Asset Pricing

## 14.1 Financial Interpretation of the Manifold

The Generalized Expenditure Manifold

$$PD = AK^{\alpha}L^{\beta}T^{\gamma}$$

admits a natural financial interpretation.

The right-hand side represents the productive foundation from which future cash flows emerge.

Define

$$Y = AK^{\alpha}L^{\beta}T^{\gamma}.$$

Then

$$PD = Y.$$

The manifold therefore links market prices and market demand directly to productive output.

## 14.2 Asset Valuation

Suppose future productive capacity evolves according to

$$Y_t = A_t K_t^{\alpha} L_t^{\beta} T_t^{\gamma}.$$

The value of an asset becomes

$$V = \sum_{t=0}^{\infty} \frac{Y_t}{(1+r)^t},$$

where $r$ denotes the discount rate.

Substituting:

$$V = \sum_{t=0}^{\infty} \frac{A_t K_t^{\alpha} L_t^{\beta} T_t^{\gamma}}{(1+r)^t}.$$

Thus financial assets represent discounted claims upon future expenditure manifolds.

### 14.3 Risk Premiums

Suppose uncertainty enters through

$$A_t, \quad K_t, \quad L_t, \quad T_t.$$

Then

$$Y_t = A_t K_t^\alpha L_t^\beta T_t^\gamma$$

becomes stochastic.

Expected valuation becomes

$$V = \sum_{t=0}^{\infty} \frac{E[Y_t]}{(1 + r + \lambda_t)^t},$$

where $\lambda_t$ denotes the risk premium.

The manifold therefore provides a structural interpretation of risk premia.

## 15 Political Economy

### 15.1 Institutions as Productivity Multipliers

The parameter $A$ captures institutional efficiency.

Examples include:

- property rights,
- legal enforcement,
- regulatory quality,
- bureaucratic competence,
- corruption control.

Decompose

$$A = A_P A_R A_B A_C,$$

where

- $A_P$ = property-rights efficiency,
- $A_R$ = regulatory efficiency,
- $A_B$ = bureaucratic efficiency,
- $A_C$ = corruption-adjustment factor.

The manifold becomes

$$PD = A_P A_R A_B A_C K^\alpha L^\beta T^\gamma.$$

## 15.2 Political Stability

Let

$$S \in [0, 1]$$

measure political stability.
Then

$$A = A_0 S.$$

The manifold becomes

$$PD = A_0 S K^\alpha L^\beta T^\gamma.$$

Instability compresses the expenditure manifold.

# 16 State Capacity

## 16.1 Definition

Define state capacity

$$\Sigma = f(\text{taxation}, \text{administration}, \text{security}, \text{information}).$$

State capacity influences

$$A.$$

Hence

$$A = A(\Sigma).$$

The manifold becomes

$$PD = A(\Sigma) K^\alpha L^\beta T^\gamma.$$

## 16.2 State Capacity Theorem

**Theorem 3.** *Holding capital, labor, and technology fixed, an increase in state capacity expands expenditure capacity.*

*Proof.* Assume

$$\frac{dA}{d\Sigma} > 0.$$

Then

$$\frac{dB}{d\Sigma} = K^\alpha L^\beta T^\gamma \frac{dA}{d\Sigma}.$$

Since all factors are positive,

$$\frac{dB}{d\Sigma} > 0.$$

Therefore the expenditure manifold expands.

□

# 17 Geopolitics

## 17.1 Geopolitical Access

Nations do not operate in isolation.

Let

$$G = G(\text{trade access}, \text{alliances}, \text{resource access}).$$

Then

$$A = A_0 G.$$

The manifold becomes

$$PD = A_0 G K^{\alpha} L^{\beta} T^{\gamma}.$$

## 17.2 Sanctions

Suppose sanctions reduce

- trade access,
- financial access,
- technological access.

Then

$$G < 1.$$

Consequently

$$B = A_0 G K^{\alpha} L^{\beta} T^{\gamma}$$

falls.

The entire expenditure manifold contracts.

# 18 International Relations

## 18.1 Alliance Networks

Suppose

$$N$$

countries form a network.

Define

$$W_{ij}$$

as the strength of ties between countries $i$ and $j$.

Then geopolitical effectiveness may be represented by

$$G_i = \sum_j W_{ij}.$$

The manifold for nation $i$ becomes

$$PD = A_i G_i K_i^{\alpha} L_i^{\beta} T_i^{\gamma}.$$

## 18.2 Strategic Dependence

Dependence on foreign technology creates

$$T = T_D + T_F,$$

where

- $T_D$ = domestic technology,
- $T_F$ = foreign technology.

Geopolitical shocks may reduce

$$T_F.$$

This directly compresses expenditure capacity.

# 19 Warfare Economics

## 19.1 Destruction Dynamics

War affects

$$K, \quad L, \quad T, \quad A.$$

Infrastructure destruction reduces

$$K.$$

Casualties reduce

$$L.$$

Institutional breakdown reduces

$$A.$$

Therefore

$$PD = AK^{\alpha}L^{\beta}T^{\gamma}$$

contracts.

## 19.2 Innovation Dynamics

Military competition often accelerates innovation.

Suppose

$$T_{t+1} = T_t + \Delta T.$$

Then technological gains partially offset losses elsewhere.

War therefore creates opposing forces:

1. destruction,
2. innovation.

## 19.3 Military Expenditure

Define

$$M_W$$

as military expenditure.
Suppose

$$T = T(M_W).$$

Then

$$\frac{dT}{dM_W} > 0.$$

The long-run effect of defense spending may therefore increase productive technology.

# 20 Strategic State Capacity

## 20.1 The Security Multiplier

Let

$$Q$$

denote security.
Then

$$A = A_0 Q.$$

The manifold becomes

$$PD = A_0 Q K^\alpha L^\beta T^\gamma.$$

## 20.2 Strategic Security Theorem

**Theorem 4.** *If productive assets are vulnerable to disruption, security acts as a multiplicative productivity factor.*

*Proof.* Without security:

$$Q < 1.$$

With complete security:

$$Q = 1.$$

Since

$$B = A_0 Q K^\alpha L^\beta T^\gamma,$$

the result follows immediately.

□

## 21 The Architecture of Nations

The manifold suggests that nations possess four foundational pillars:

1. Capital.
2. Labor.
3. Technology.
4. Institutions.

The resulting expenditure capacity is

$$B = AK^{\alpha}L^{\beta}T^{\gamma}.$$

Political and strategic systems exist largely to preserve and expand these pillars.

## 22 A Geopolitical Diagram

$$\begin{tikzcd}
K \arrow[dr] & & & \\
L \arrow[r] & A \arrow[r] & B \arrow[r] & PD \\
T \arrow[ur] & & &
\end{tikzcd}$$

Figure 2: Production factors generating expenditure capacity.

## 23 Comparative Political-Economic Systems

| System | Capital | Institutions | Expected $A$ |
|---|---|---|---|
| Market Democracy | High | Strong | High |
| Developmental State | High | Strong | High |
| Failed State | Low | Weak | Low |
| War Economy | Variable | Variable | Variable |
| Command Economy | Moderate | Centralized | Mixed |

Table 1: Illustrative political-economic comparisons.

## 24 Implications

The generalized framework implies:

1. Demand cannot be understood independently of production.
2. Production cannot be understood independently of institutions.
3. Institutions cannot be understood independently of state capacity.
4. State capacity cannot be understood independently of security.
5. Security cannot be understood independently of geopolitics.

Thus the expenditure manifold links microeconomics to grand strategy.

# 25 Statistical Foundations

## 25.1 Observable Variables

Suppose observations are available for

$$P_t, \quad D_t, \quad K_t, \quad L_t, \quad T_t, \quad A_t.$$

The Generalized Expenditure Manifold implies

$$P_t D_t = A_t K_t^{\alpha} L_t^{\beta} T_t^{\gamma}.$$

Define

$$Y_t = P_t D_t.$$

Then

$$Y_t = A_t K_t^{\alpha} L_t^{\beta} T_t^{\gamma}.$$

This converts the expenditure manifold into a production-like statistical system.

## 25.2 Logarithmic Form

Taking logarithms:

$$\ln Y_t = \ln A_t + \alpha \ln K_t + \beta \ln L_t + \gamma \ln T_t.$$

Introducing disturbances:

$$\ln Y_t = a + \alpha \ln K_t + \beta \ln L_t + \gamma \ln T_t + \varepsilon_t.$$

This specification is directly estimable by ordinary least squares.

# 26 Econometric Estimation

## 26.1 Linear Regression

The baseline regression becomes

$$y_t = a + \alpha k_t + \beta l_t + \gamma t_t + \varepsilon_t,$$

where

$$y_t = \ln(P_t D_t).$$

Parameter estimates provide direct estimates of manifold elasticities.

## 26.2 Elasticity Interpretation

The coefficients admit immediate interpretation:

$$\alpha = \frac{\partial \ln B}{\partial \ln K},$$

$$\beta = \frac{\partial \ln B}{\partial \ln L},$$

$$\gamma = \frac{\partial \ln B}{\partial \ln T}.$$

Thus the manifold provides a direct bridge between production economics and expenditure economics.

# 27 Hypothesis Testing

## 27.1 Null Hypothesis

The fundamental hypothesis is

$$H_0 : PD = AK^\alpha L^\beta T^\gamma.$$

The alternative is

$$H_1 : PD \neq AK^\alpha L^\beta T^\gamma.$$

## 27.2 Likelihood Framework

Assume

$$\varepsilon_t \sim N(0, \sigma^2).$$

Then

$$\ln L = -\frac{n}{2} \ln(2\pi\sigma^2) \frac{1}{2\sigma^2} \sum_t \varepsilon_t^2.$$

Maximum likelihood estimation follows immediately.

# 28 Structural Breaks

## 28.1 Motivation

Wars, sanctions, technological revolutions, and political transitions may alter

$$A, \quad \alpha, \quad \beta, \quad \gamma.$$

Therefore the manifold may experience structural breaks.

## 28.2 Regime Specification

Let

$$S_t \in 1, 2, \ldots, m.$$

Then

$$PD = A_{S_t} K^{\alpha_{S_t}} L^{\beta_{S_t}} T^{\gamma_{S_t}}.$$

Each regime corresponds to a different expenditure manifold.

# 29 Bayesian Formulation

Treat

$$A, \alpha, \beta, \gamma$$

as random variables.
Prior distributions:

$$A \sim p(A),$$

$$\alpha \sim p(\alpha),$$

$$\beta \sim p(\beta),$$

$$\gamma \sim p(\gamma).$$

Posterior inference follows:

$$p(\theta|X) = \frac{p(X|\theta)p(\theta)}{p(X)},$$

where

$$\theta = (A, \alpha, \beta, \gamma).$$

# 30 Data Science Architecture

## 30.1 Data Pipeline

The manifold requires:

1. Price data.
2. Demand data.
3. Capital-stock data.
4. Labor-force data.
5. Technology indicators.
6. Institutional indicators.

A generalized pipeline is

$$\text{Raw Data} \to \text{Cleaning} \to \text{Feature Engineering} \to \text{Estimation} \to \text{Forecasting}.$$

## 30.2 Feature Engineering

Construct:

$$Y_t = P_t D_t.$$

Additional features include

$$\ln K_t, \quad \ln L_t, \quad \ln T_t.$$

Institutional indicators may include

- governance scores,
- corruption measures,
- security measures,
- state-capacity measures.

# 31 Machine Learning

## 31.1 Structural Prior

The manifold may be treated as a structural prior:

$$PD = AK^{\alpha}L^{\beta}T^{\gamma}.$$

Machine learning models then estimate residual variation.

## 31.2 Residual Learning

Suppose

$$\widehat{B} = AK^{\alpha}L^{\beta}T^{\gamma}.$$

Define

$$u_t = P_t D_t - \widehat{B}_t.$$

Neural networks estimate

$$u_t = f(X_t).$$

The final model becomes

$$P_t D_t = AK^{\alpha}L^{\beta}T^{\gamma} + f(X_t).$$

## 31.3 Hybrid Intelligence

The resulting architecture combines:

1. Economic theory.
2. Statistical inference.
3. Machine learning.

This often outperforms purely statistical systems.

# 32 Artificial Intelligence

## 32.1 State Representation

Define state vector

$$x_t = (K_t, L_t, T_t, A_t).$$

The expenditure manifold determines

$$B_t.$$

The AI agent observes

$$x_t.$$

## 32.2 Control Variables

Control variables include

$$u_t = (M_t, F_t, R_t),$$

where

- $M_t$ = monetary policy,
- $F_t$ = fiscal policy,
- $R_t$ = regulatory policy.

The control problem becomes

$$\max E\left[\sum_t \delta^t U_t\right].$$

# 33 Reinforcement Learning

State transition:

$$x_{t+1} = g(x_t, u_t).$$

Reward:

$$R_t = w_1 B_t + w_2 D_t - w_3 P_t.$$

Policy:

$$\pi(u_t|x_t).$$

The expenditure manifold supplies the environment dynamics.

# 34 Computational Complexity

## 34.1 Direct Evaluation

Evaluation of

$$B = AK^{\alpha} L^{\beta} T^{\gamma}$$

requires constant time.
Therefore

$$O(1).$$

## 34.2 Panel Estimation

For

$$N$$

countries and

$$T$$

periods,
complexity becomes

$$O(NT).$$

The framework therefore scales efficiently.

# 35 Algorithms

## 35.1 Manifold Evaluation

```
Input:
A,K,L,T,alpha,beta,gamma

Output:
B

Procedure:
B = A*K^alpha*L^beta*T^gamma
Return B
```

### 35.2 Price Estimation

```
Input:
B,D

Output:
P

Procedure:
P = B/D
Return P
```

### 35.3 Demand Estimation

```
Input:
B,P

Output:
D

Procedure:
D = B/P
Return D
```

## 36 Forecasting

### 36.1 Dynamic Technology

Suppose

$$T_{t+1} = T_t + \Delta T_t.$$

Then

$$B_{t+1} = A_{t+1} K_{t+1}^{\alpha} L_{t+1}^{\beta} T_{t+1}^{\gamma}.$$

Forecasting expenditure capacity becomes a forecasting problem for productive factors.

### 36.2 Long-Run Growth

Differentiating logarithmically:

$$\frac{\dot{B}}{B} = \frac{\dot{A}}{A} + \alpha\frac{\dot{K}}{K} + \beta\frac{\dot{L}}{L} + \gamma\frac{\dot{T}}{T}.$$

This expression links manifold growth directly to growth theory.

## 37 A Computational Diagram

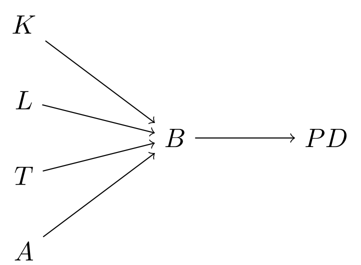


Figure 3: Computational architecture of the expenditure manifold.

# 38 Comparison with Classical Demand Theory

| Feature | Linear Demand | Expenditure Manifold |
|---|---|---|
| Geometry | Linear | Hyperbolic |
| Elasticity | Variable | Constant |
| Production Link | Weak | Explicit |
| Technology | Absent | Explicit |
| Institutions | Absent | Explicit |
| State Capacity | Absent | Explicit |
| Geopolitics | Absent | Explicit |
| Warfare | Absent | Explicit |
| AI Integration | Weak | Natural |

Table 2: Comparison of frameworks.

# 39 Generalized Expenditure Manifold Principle

**Theorem 5.** *Every economically realizable price-demand pair must lie on a manifold generated by productive capacity.*

*Proof.* By construction,

$$B = AK^{\alpha}L^{\beta}T^{\gamma}.$$

Since

$$PD = B,$$

every feasible pair satisfies

$$PD = AK^{\alpha}L^{\beta}T^{\gamma}.$$

Hence all realizable observations lie on the manifold.

□

# 40 Final Conclusion

The Generalized Expenditure Manifold

$$PD = AK^{\alpha}L^{\beta}T^{\gamma}$$

provides a unifying framework connecting:

1. Microeconomics.
2. Macroeconomics.
3. Finance.
4. Political economy.
5. Psychology.
6. Geopolitics.
7. International relations.
8. Warfare economics.
9. Statistics.
10. Data science.

11. Artificial intelligence.
12. Computer science.

The traditional linear demand curve emerges naturally as a local tangent approximation to the manifold.

Production, institutions, technology, and state capacity jointly determine expenditure capacity, while monetary and fiscal policy influence movement across and within the manifold.

The resulting framework replaces isolated demand curves with a higher-dimensional geometry of economic possibility.

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## Glossary

$A$ Institutional efficiency parameter.

$B$ Expenditure capacity.

$D$ Demand.

$K$ Capital stock.

$L$ Labor input.

$P$ Price.

$T$ Technology.

$\alpha$ Capital elasticity.

$\beta$ Labor elasticity.

$\gamma$ Technology elasticity.

**Expenditure Manifold** The surface satisfying

$$PD = AK^{\alpha}L^{\beta}T^{\gamma}.$$

**Boundary Equilibrium** Feasible endpoint of a manifold arc.

**Institutional Efficiency** Productivity contribution arising from governance and state capacity.

**State Capacity** Administrative and security capability of a political system.

**Tangent Demand Curve** Local linear approximation to the manifold.

## The End